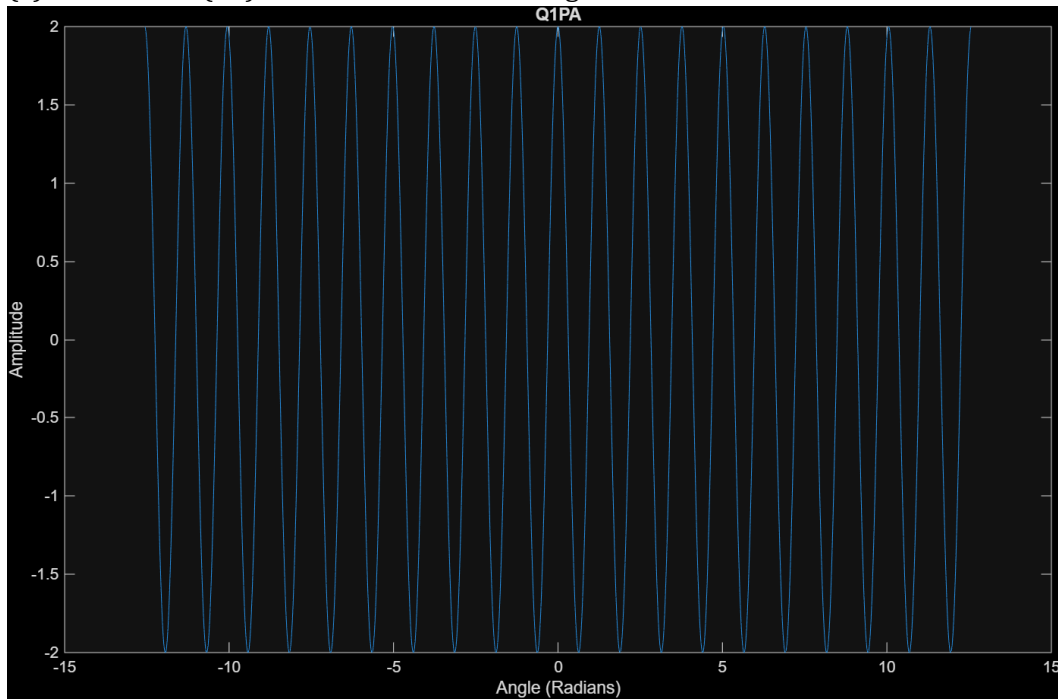


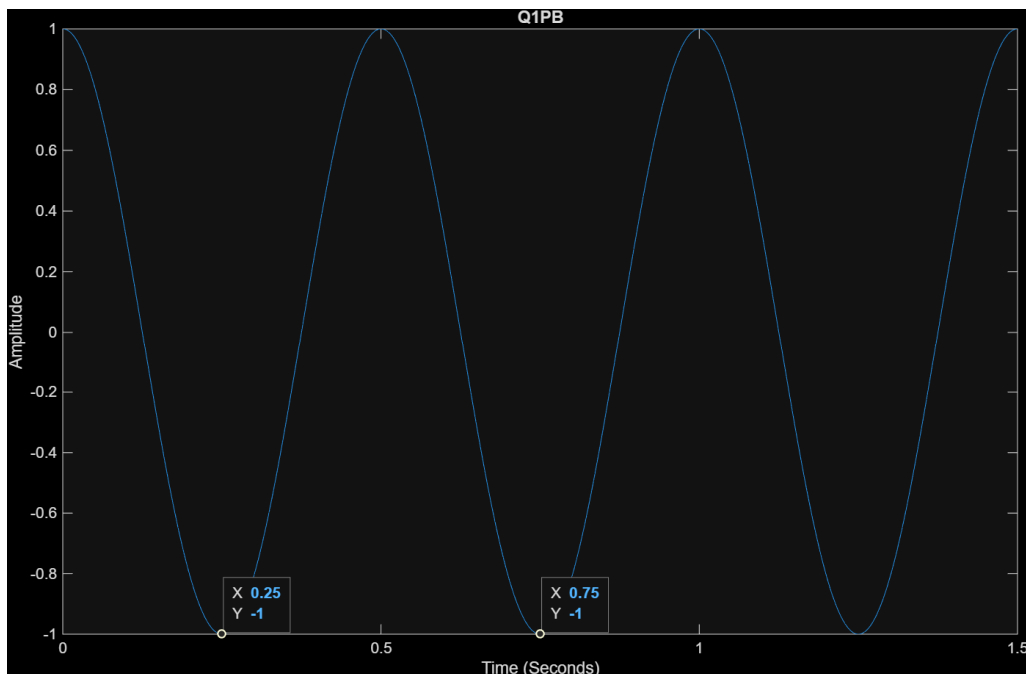
Full Name: Evan Baesler
EEL 3135 (Fall 2025) – HW #01
Homework 01

HW Question #1: Use matlab to plot the following signals using the Cosine function:

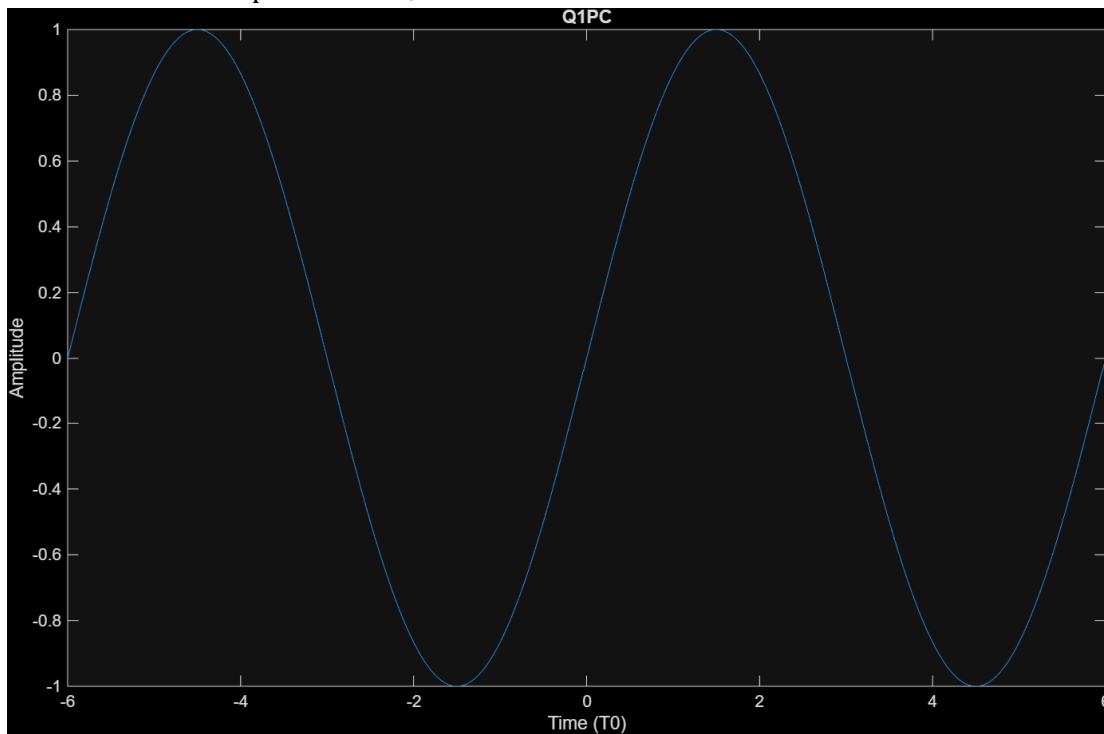
(a) Sketch $2\cos(5\theta)$ for values of θ in the range $-4\pi \leq \theta \leq 4\pi$



(b) Sketch $\cos(4\pi t)$ for values of t such that three full periods of the function are shown. Label the periods clearly.



(c) Sketch $\cos(2\pi t/T_0 - \pi/2)$ for values of t such that $-T_0 \leq t \leq T_0$, with $T_0 = 6$. Label the horizontal axis in terms of the parameter T_0 .



Use MATLAB for each case, include code and plots, and briefly discuss differences in amplitude, frequency, and phase effects.

For the amplitude, this is based on the coefficient of our cosine waves, in order it would be 2A, 1A, 1A amplitude. The frequency is determined by the coefficient to t in our waves.

For frequency: $f_a = \frac{5}{2\pi} \text{ Hz}$, $f_b = \frac{4\pi}{2\pi} = 2 \text{ Hz}$, $f_c = \frac{1}{T_0} = \frac{1}{6}$

For phase: a and b had 0 phase, where c had a phase shift of $-\frac{\pi}{2}$, shifting it to the right.

Q1 CODE:

% Part 1

% Sketch $2 \cos(5\theta)$ for values of θ in the range $-4\pi \leq \theta \leq 4\pi$

```
clear all;  
close all;  
clc;
```

```
figure;
```

```
A = 2;  
fs = 1000;  
dt = 1/fs;  
theta = -4*pi:dt:(4*pi);  
y = A*cos(5*theta);
```

```
plot(theta, y)  
ylabel('Amplitude')  
xlabel('Angle (Radians)')  
title('Q1PA');
```

% Part 2

% Sketch $\cos(4\pi t)$ for values of t such that three full periods of the function are shown.
% Label the periods clearly.

```
figure;
```

```
A = 1;  
fs = 1000;  
dt = 1/fs;  
t = 0*pi:dt:1.5; %  $T = 2\pi/w = (2\pi)/(4\pi) = 1/2$   
y = A*cos(4*pi*t);
```

```
plot(t, y)  
ylabel('Amplitude')  
xlabel('Time (Seconds)')  
title('Q1PB')
```

% Part 3

% Sketch $\cos(2\pi t/T_0 - \pi/2)$ for values of t such that $-T_0 \leq t \leq T_0$, with $T_0 = 6$.

% Label the horizontal axis in terms of the parameter T_0 .

figure;

A = 1;

fs = 100;

dt = 1/fs;

T0 = 6;

t = -T0:dt:T0; % $T = 2\pi/w = (2\pi)/(4\pi) = 1/2$

y = A*cos(((2*pi*t)/T0)-pi/2);

plot(t, y)

ylabel('Amplitude')

xlabel('Time (T0)')

title('Q1PC')

HW Question #2: Reconstructing a Sinusoid A sinusoid has the following properties:

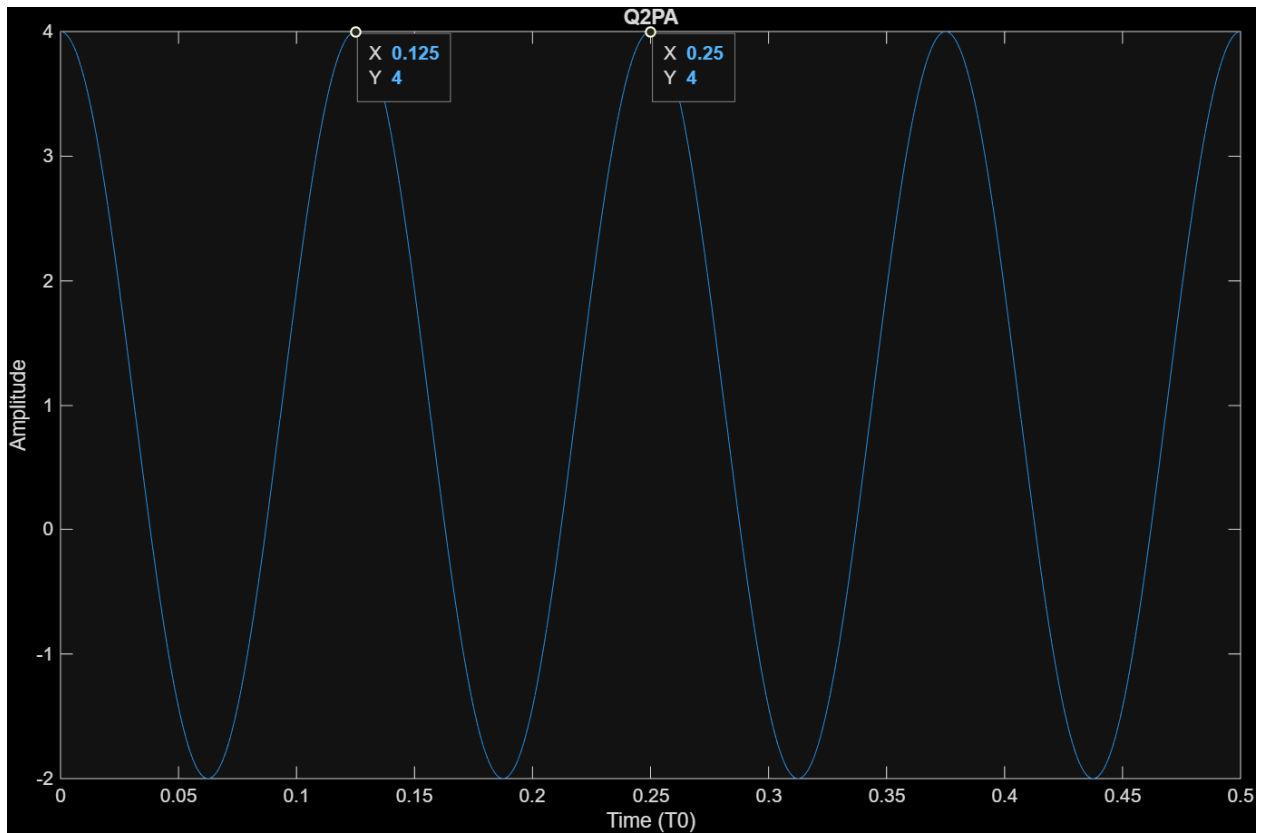
- Amplitude = 3
- Frequency = 8 Hz
- DC offset = +1
- At $t=0$, the signal equals its maximum value.

(a) Write the function $x(t)$ in the form

$$x(t) = A \cos(\omega_0 t + \varphi) + D.$$

$$x(t) = 3 \cos(8 * 2\pi * t + 0) + 1 = \mathbf{3 \cos(16\pi t) + 1}$$

(b) Plot the signal in MATLAB for $0 \leq t \leq 0.5$ seconds.



(c) Discuss how amplitude, frequency, and phase reveal themselves in your plot.

The wave goes from -2 to 4, which means we have an amplitude of 3 $\rightarrow [(4 - -2)/2]$. The frequency is $1/T$ which is $\frac{1}{0.125} = 8\text{Hz}$, and a phase of 0 means we start at $\cos(0)$ or 1, $1 * 3 + 1 = 4$.

Q2 CODE:

% A sinusoid has the following properties:

% • Amplitude = 3

% • Frequency = 8 Hz

% • DC offset = +1

% • At $t=0$, the signal equals its maximum value.

% Part 1

% Write the function $x(t)$ in the form: $x(t) = A \cos(\omega_0 t + \phi) + D$.

figure;

fs = 1000;

dt = 1/fs;

A = 3;

f = 8;

$\omega_0 = 2\pi f$;

D = 1;

t = 0:dt:0.5;

Phi = 0;

$y = A \cos(\omega_0 t + \text{Phi}) + D$;

% Part 2

% Plot the signal in MATLAB for $0 \leq t \leq 0.5$ seconds.

plot(t, y)

ylabel('Amplitude')

xlabel('Time (T0)')

title('Q2PA')

% Part 3

% Discuss how amplitude, frequency, and phase reveal themselves in your plot.

% Amplitude shows the maximum absolute value of the peaks/mins of the wave.

% Frequency is revealed because $1/8 = 0.125$ seconds, our gap between peaks.

% Our phase is zero, so our cosine wave has $\cos(0) = 1A + D$, or $3+1 = 4$.

HW Question #3: Two phasors are defined as:

$$z_A = 3 + 4j \quad z_B = -2 + 5j$$

rotating at 5Hz.

Complete the following using calculations by hand:

(a) Plot z_A and z_B using MATLAB's compass function. **Image of all phasors below**

(b) Find $z_S = z_A + z_B$.

$$z_S = 3 - 2 + 4j + 5j = \mathbf{1 + 9j \text{ (Rectangular)}}$$

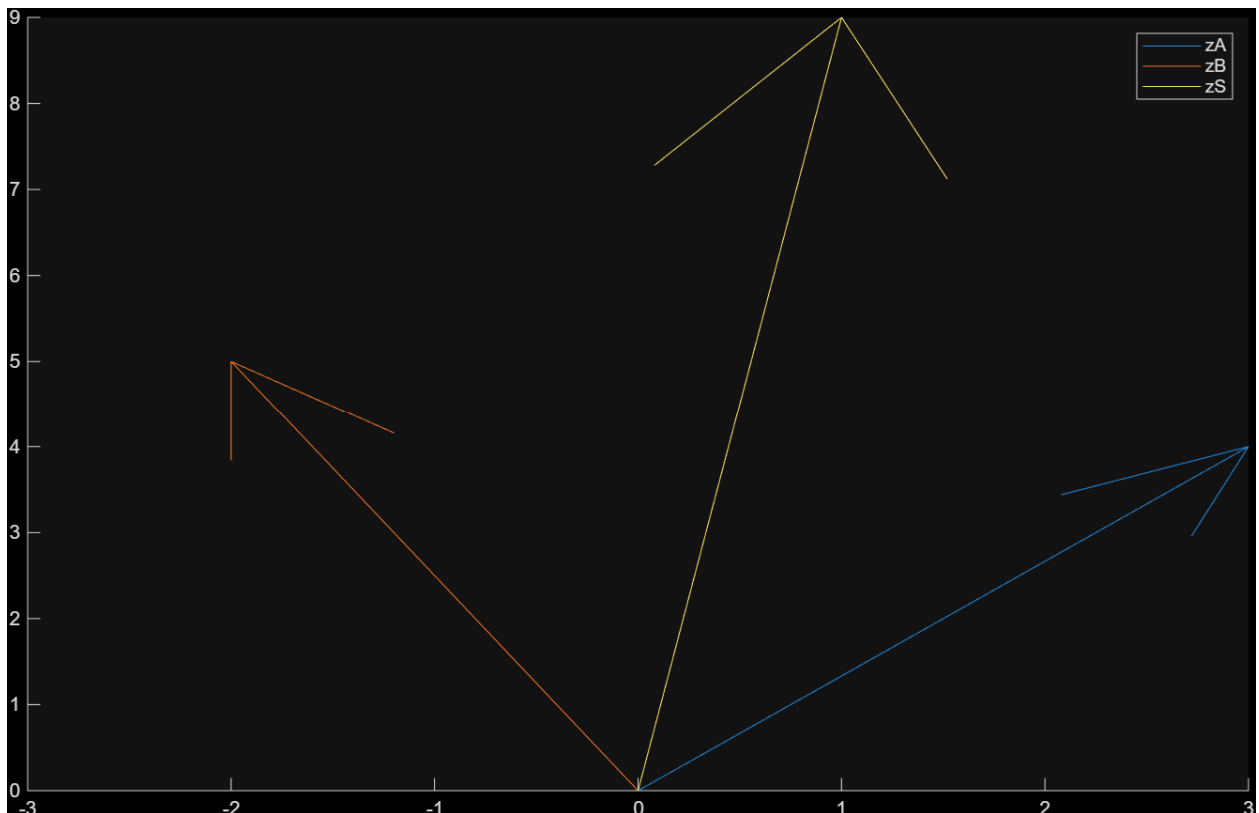
(c) Express in rectangular and polar form z_S

$$|z_S| = \sqrt{1^2 + 9^2} = \sqrt{82} \cong 9.055, \angle = \arctan\left(\frac{9}{1}\right) \cong 1.46 \frac{\text{rads}}{s} \rightarrow z_S = \mathbf{9.055 \angle 1.46 \frac{\text{rads}}{s} \text{ (Polar)}}$$

(d) Now plot $z_S = z_A + z_B$ using MATLAB and check your work. **Image of all phasors below**

(e) Analyze z_S , what is the cosine function the would describe it?

$$Y = |z_S| * \cos(\omega_0 t + \phi) = \mathbf{9.055 \cos(31.416t + 1.460) \text{ (in } \frac{\text{rads}}{s})}$$



Q3 CODE:

% Part 1

$z_A = 3 + 4j$;

$z_B = -2 + 5j$;

% Rotating at 5 Hz

figure;

hold on;

$f = 5$;

compass(z_A)

compass(z_B)

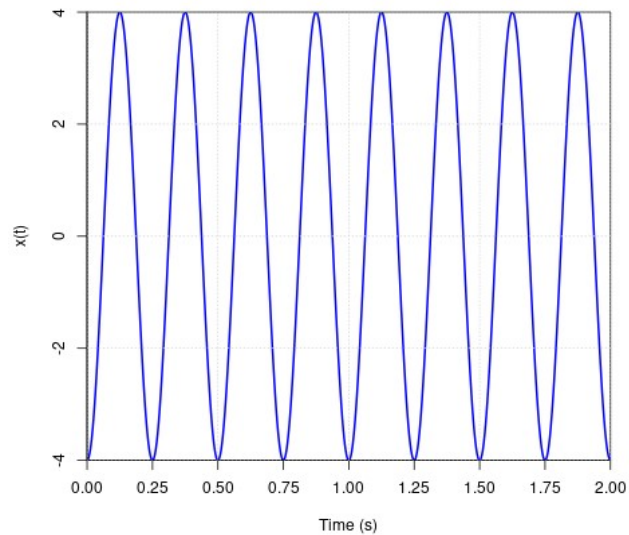
$z_S = z_A + z_B$; % This is $3-2 + 4j + 5j = 1 + 9j$ in rectangular.

compass(z_S)

legend

hold off;

HW Question #4: Complex Sinusoid Representation



- (a) Find the equation for the complex sinusoid. Write the function $x(t)$ in the form

$$x(t) = A \cos(\omega_0 t + \varphi) + D.$$

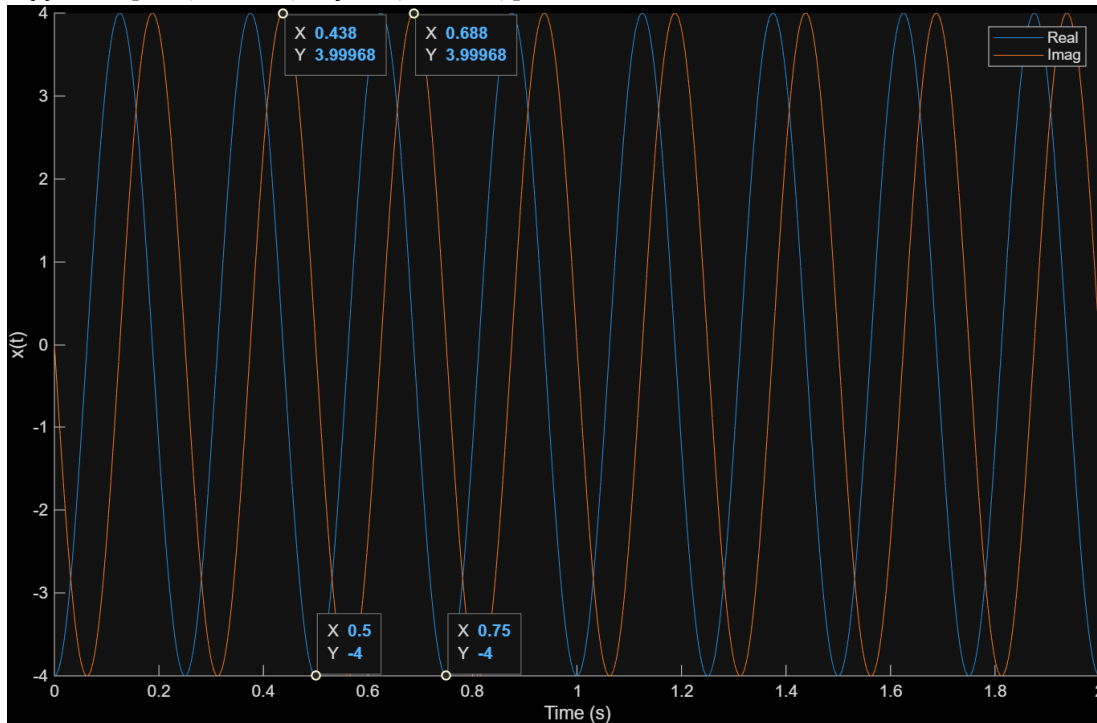
$$x(t) = 4 \cos(8\pi t \pm \pi) + 0$$

- (b) For the sinusoidal waveform shown, determine the complex phasor representation $X = Ae^{j\varphi}$ by finding the values of the parameters ω_0 , φ , and A such that the waveform is represented by

$$x(t) = \mathcal{R} \{ X e^{j\omega_0 t} \} = \mathcal{R} \{ 4e^{j8\pi t + \pi} \}$$

- (c) Plot real and imaginary component of the function.

$$Y(t) = 4 * [\cos(8\pi t + \pi) + j \sin(8\pi t + \pi)]$$



Q4 CODE:

% Part 3

```
fs = 1000;  
dt = 1/fs;
```

```
A = 4;  
f = 4;  
W0 = 2*f*pi;  
t = 0:dt:2;  
Phi = pi;
```

```
figure;  
hold on;
```

```
realcomp = 4*cos(W0*t+pi)  
imagcomp = 4*sin(W0*t+pi)
```

```
plot(t, realcomp)  
plot(t, imagcomp)  
ylabel('x(t)')  
xlabel('Time (s)')  
legend;
```

```
hold off;
```

HW Question #5: Simplify the following sum:

$$5\cos(40\pi t + \pi/6) + 2\cos(40\pi t - \pi/3) \quad (1)$$

- (a) Add the two sinusoids and express it in form of $A\cos(2\pi ft + \varphi)$ using the complex sinusoids formula ($e^{j\theta}$).

$$5\cos\left(40\pi t + \frac{\pi}{6}\right) = 5e^{j\frac{\pi}{6}} \mid 2\cos\left(40\pi t - \frac{\pi}{3}\right) = 2e^{-j\frac{\pi}{3}}$$

This gives us:

$$5e^{j\frac{\pi}{6}} = 5\cos\left(\frac{\pi}{6}\right) + 5j\sin\left(\frac{\pi}{6}\right) \text{ and } 2e^{-j\frac{\pi}{3}} = 2\cos\left(-\frac{\pi}{3}\right) + 2j\sin\left(-\frac{\pi}{3}\right)$$

Which is equivalent to:

$$5e^{j\frac{\pi}{6}} = 5\left(\frac{\sqrt{3}}{2}\right) + 5\left(j\frac{1}{2}\right) = \frac{5\sqrt{3}}{2} + j\frac{5}{2} \text{ and } 2e^{-j\frac{\pi}{3}} = 2\left(\frac{1}{2}\right) - 2\left(j\frac{\sqrt{3}}{2}\right) = 1 - j\sqrt{3}$$

Simplified:

$$\frac{5\sqrt{3}}{2} + 1 + j\frac{5}{2} - j\sqrt{3} \cong 5.330 + j0.768$$

Magnitude:

$$\sqrt{(5.33)^2 + (0.768)^2} \cong 5.385$$

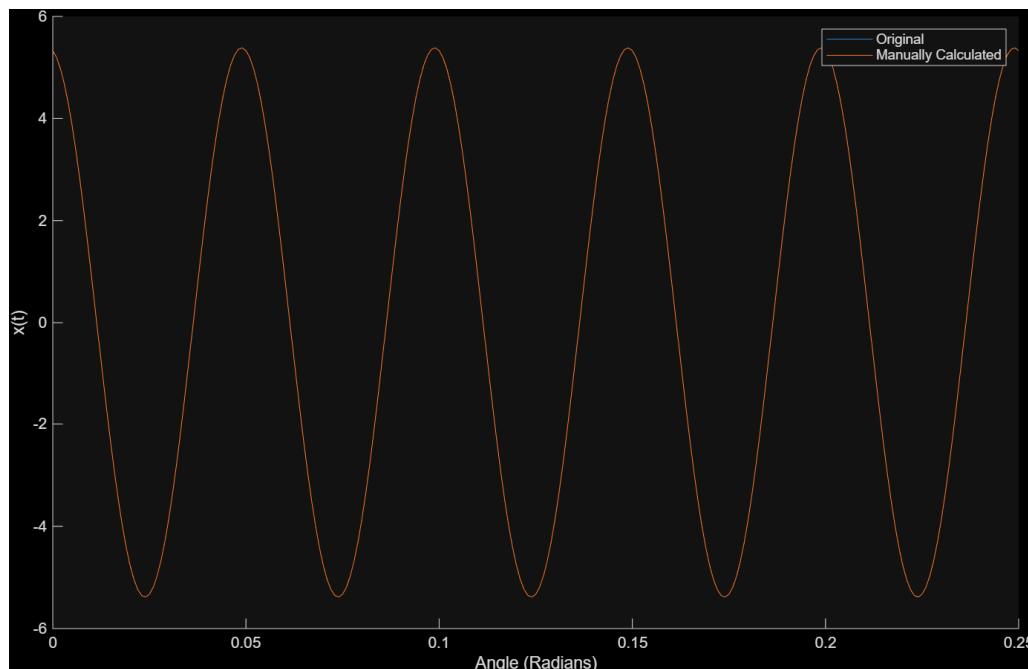
Angle:

$$\arctan\left(\frac{0.768}{5.330}\right) \cong 0.143 \text{ rad}$$

Final:

$$5.385\cos(2\pi * 20 * t + 0.143)$$

- (b) Verify your result in MATLAB by plotting both the original sum and your simplified expression on the same graph for time t between 0 and 0.25 seconds.



Q5 CODE:

```
% Part 2
```

```
fs = 1000;
```

```
dt = 1/fs;
```

```
A = 5.385;
```

```
f = 20;
```

```
W0 = 2*f*pi;
```

```
t = 0:dt:0.25;
```

```
Phi = 0.143;
```

```
waveSum1 = (5*cos(40*pi*t+(pi/6)) + 2*cos(40*pi*t - (pi/3)))
```

```
figure;
```

```
hold on;
```

```
plot(t, waveSum1)
```

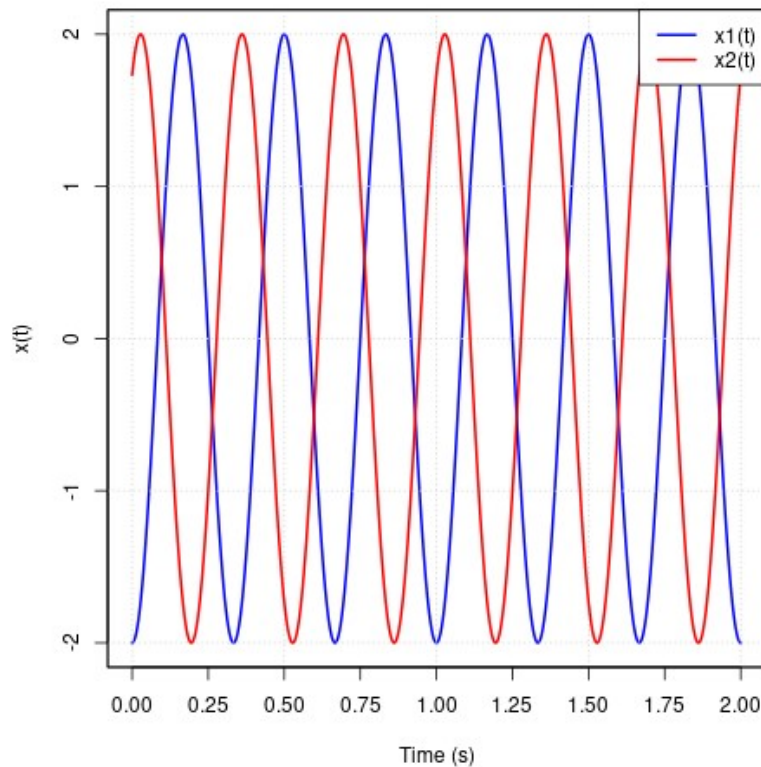
```
plot(t, A*cos(W0*t+Phi))
```

```
ylabel('x(t)')
```

```
xlabel('Angle (Radians)')
```

```
legend('Original', 'Manually Calculated');
```

HW Question #6: Consider the following graph:



- (a) Find the equation of the two complex sinusoids. Write the functions $x_1(t)$ and $x_2(t)$ in the form $x(t) = A\cos(\omega_0 t + \varphi) + D$.

$$x_1(t) = 2 \cos(12\pi t), x_2(t) = 2 \cos(12\pi t + \pi)$$

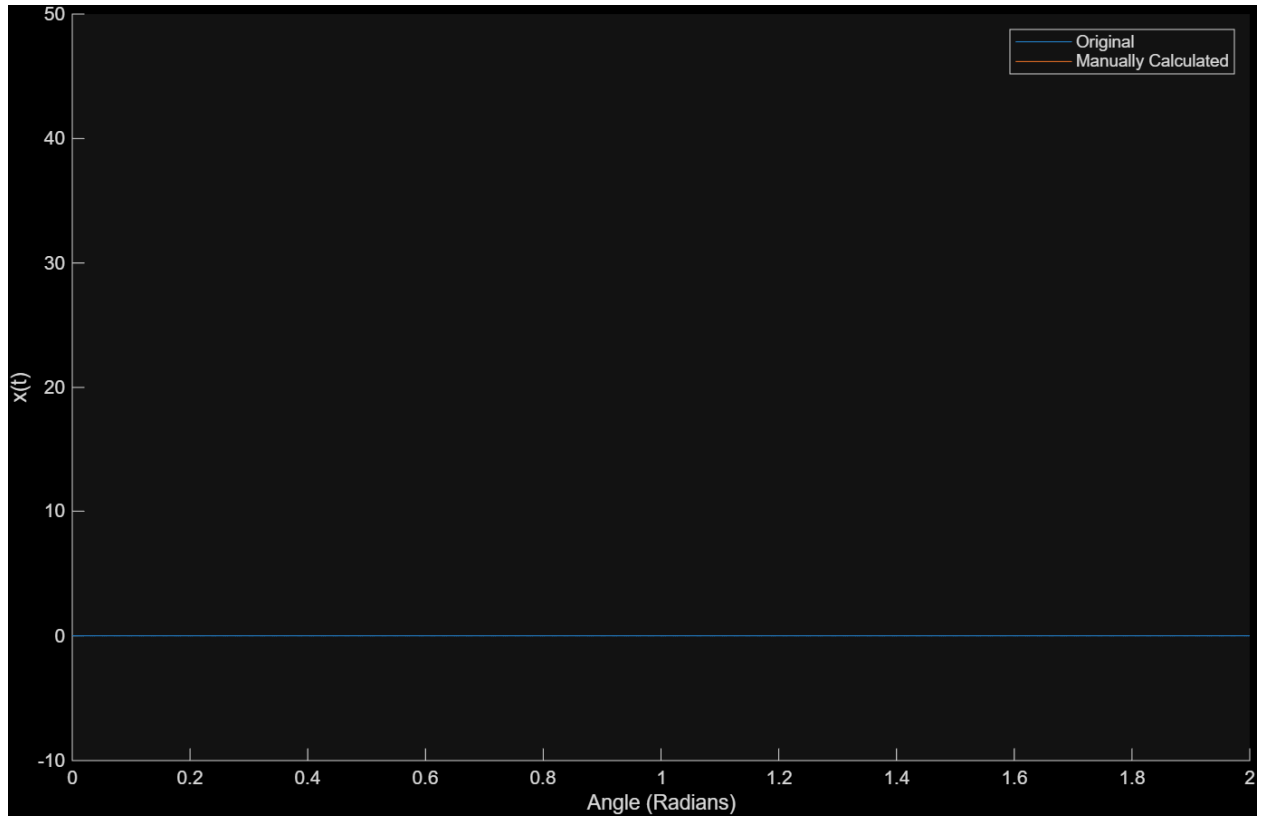
- (b) Add the two sinusoids $x_1(t)$ and $x_2(t)$, and express it in the form of $A\cos(2\pi f t + \varphi)$ using the complex sinusoids formula ($e^{j\theta}$).

$$x_1(t) = 2 \left(\frac{e^{j12\pi t} + e^{-j12\pi t}}{2} \right) = (e^{j12\pi t} + e^{-j12\pi t})$$

$$x_2(t) = 2 \frac{(e^{j12\pi t + \pi} + e^{-j12\pi t + \pi})}{2} = e^{j12\pi t + \pi} + e^{-j12\pi t + \pi}$$

Since they are offset by pi, the terms will cancel out, positive with positive and negative with negative. Net sum of 0; $x_1(t) + x_2(t) = 0$

- (c) Now plot the sinusoids from part a and part b in MATLAB. You should have three sinusoids, $x_1(t)$ and $x_2(t)$, and the summation of both $x_1(t)$ and $x_2(t)$. In 2 to 3 sentences, explain how the addition of $x_1(t)$ and $x_2(t)$ leads to the construction or deconstruction of the resulting summation sinusoid (HINT: look at the phase).



The two waves are offset from each other by 1 pi, causing them to cancel one another out. This is deconstruction, as we start with two distinct waves/data and end with a flatline. This happens because the term $e^{j12\pi}$ is canceled out by $e^{-j12\pi}$ and $e^{j12\pi+\pi}$ is canceled out by $e^{-j12\pi+\pi}$.

Q6 CODE:

```
% Part 3
```

```
fs = 1000;  
dt = 1/fs;
```

```
A = 2;  
f = 6;  
W0 = 2*f*pi;  
t = 0:dt:2;
```

```
waveSum2 = (2*cos(12*pi*t) + 2*cos(12*pi*t + (pi)))
```

```
figure;  
hold on;
```

```
plot(t, waveSum2)  
plot(t, (exp(j*12*pi) + exp(-1*j*12*pi) + exp(j*12*pi + pi) + exp(-1*j*12*pi + pi)))  
ylabel('x(t)')  
xlabel('Angle (Radians)')  
legend('Original', 'Manually Calculated');
```